

Parmar Jyotiraditya Kamleshkumar

AI & ML / Generative AI Engineer

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AI/ML Engineer specializing in Generative AI, autonomous agents, and production-ready RAG architectures. Experienced in building multi-agent workflow systems (LangGraph), real-time streaming tools (FastAPI, WebSockets), and high-performance vector databases. Passionate about developer tooling, open-source development, and creating end-to-end data intelligence applications.

EXPERIENCE

Machine Learning Intern

Jun 2024 – Jul 2024

Cognifyz Technologies

Remote

- Implemented supervised and unsupervised ML algorithms (regression, classification, clustering) on real-world datasets, applying grid search hyperparameter tuning and cross-validation to improve baseline accuracy by 14%.
- Built modular ETL data pipelines for data lake ingestion and preprocessing, exposing model inference workflows via robust FastAPI REST endpoints for real-time application consumption.
- Conducted comprehensive statistical analysis and data cleaning on high-dimensional feature spaces, identifying key correlations and reducing input noise to optimize overall training performance.

PROJECTS

LuminateAI — Enterprise RAG & Data Analytics Platform

GitHub

LangGraph · FastAPI · Next.js · ChromaDB · Qdrant · Cohere Rerank · Docker · WebSockets · SQL

- Designed and implemented a multi-agent RAG workflow using LangGraph to ingest **10k+ pages** of unstructured documents (PDFs, spreadsheets) and automatically generate structured analytical summaries with **92% retrieval precision**.
- Built a custom retrieval pipeline with hybrid search (dense semantic embeddings + sparse BM25) coupled with Reciprocal Rank Fusion (RRF) and rerankers, achieving a **95% accuracy rate** on domain-specific queries.
- Developed an interactive analytics dashboard in React/Next.js that translates natural language queries into SQL and pandas commands, reducing query generation time by **60% (from 5s to sub-2s)** with WebSocket streaming.
- Implemented context-aware query routing using a router LLM that directs transactional queries to PostgreSQL and unstructured queries to vector databases, improving throughput by **35%**.

RepoMind — AI-Powered Codebase Explorer & Chat Assistant

GitHub

FastAPI · LangGraph · Qdrant · AST Parsers · OpenAI API · Git · Docker · Python

- Engineered an agentic chat assistant that indexes large GitHub repositories (up to **500k+ LOC**), building hierarchical semantic maps of codebases using AST parsers and tree-structured vector indexes.
- Implemented an autonomous debugging and refactoring loop in LangGraph that reads issue descriptions, locates target files, and generates correct git patches with a **78% first-pass compiler success rate**.
- Optimized LLM context window management through dynamic code chunking, repository structure summaries, and dependency tracing, reducing token overhead by **45%** and shortening agent response latency by **35%**.
- Designed a self-correcting agent loop that runs tests in sandboxed Docker containers, automatically fixing compiler and runtime errors to increase final patch correctness by **22%**.

macspot — Real-Time Network Discovery Agent

GitHub

Python · Scapy · Ollama · FastAPI · WebSockets · React · SQLite · CVE Database API

- Developed a network device discovery tool that executes high-frequency ARP scanning using Scapy, discovering **100+ active network nodes** in **under 12 seconds** with **99.8% detection reliability**.
- Integrated local LLMs (via Ollama) to analyze device signatures, active ports, and network behaviors to intelligently identify hardware vendors and OS operating systems with **94% classification accuracy**.
- Designed a bidirectional WebSocket communication channel in FastAPI to stream real-time scan results to a React UI, reducing UI rendering lag by **70%** while triggering concurrent CVE lookups in **under 1.5s**.
- Built a persistent local caching mechanism using SQLite to cache device details and history, accelerating subsequent scans and decreasing lookup latency by **80%**.

LEADERSHIP & ACTIVITIES

Placement Coordinator — TPO, LDCE

2024 – Present

- Coordinate campus placement drives, streamline communication between corporate recruiters and 500+ students, and organize mock interviews.

President — Innovator Club, LDCE

2024 – Present

- Lead a 60+ member student club, organizing hands-on AI/ML workshops, hackathons (200+ participants), and technical bootcamps to foster innovation.

TECHNICAL SKILLS

GenAI Stack: LangGraph, LangChain, LlamaIndex, Gemini API, RAG, Ollama, Vector Databases (Qdrant, ChromaDB, FAISS), AST Parsing

Languages: Python, TypeScript, SQL, Java, Go, C++, HTML/CSS

Frameworks: FastAPI, PyTorch, TensorFlow, React, Next.js, TailwindCSS

Tools & DBs: Git, GitHub, Docker, GCP, Azure, Scapy, WebSockets, PostgreSQL, MongoDB, SQLite

EDUCATION

Examination / Degree	University / Board	Institute	Year	CPI / %
B.E. (AI & ML)	GTU	L.D. College of Engineering, Ahmedabad	2023 – 2027	7.92 / 10